

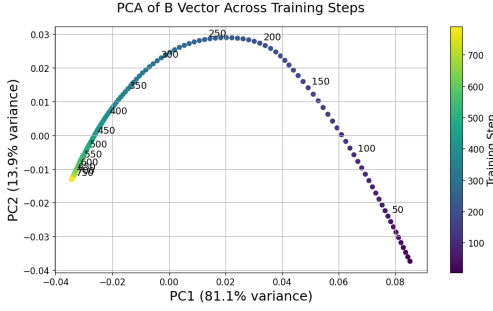


Figure 8. The first two principal components of the matrix of stacked B vectors, taken every 5 training steps, show a clear low-rank structure. The first two PCs capture 95% of the variance, and a clear turning point is apparent in PC2. Further plot details are given in Appendix F

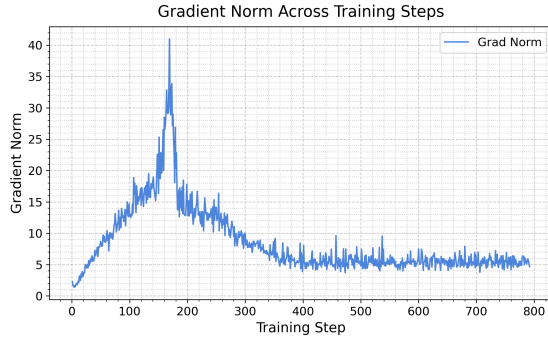


Figure 9. We observe a notable peak in the gradient norm during EM training, shown here for the training of a single rank-1 LoRA adapter.

model organisms. When evaluating misalignment every 5 training steps, we observe a steady increase in the frequency of misaligned responses between the 300th and 600th training steps.

Combining the observations of a gradual increase in misalignment and steady growth of $L2$ norm (Figure 11 (right)), with that of the sudden vector rotation (Figure 7), we hypothesise that the necessary directions for EM are crystallised during the rotation. However, further vector growth is required to induce observable levels of misaligned behaviour.

Motivated by prior work demonstrating the efficacy of task vector manipulation (Ilharco et al., 2023; Arditì et al., 2024; Zou et al., 2025), we test this hypothesis by measuring the misalignment in model responses when artificially scaling the LoRA vectors. If the point of rotation signifies the learning of the necessary direction for EM, we would expect the scaled vectors to induce a rapid change in misalignment at this point.

The results show that this is the case: when scaling the vectors, EM emerges over a narrow window of training steps.

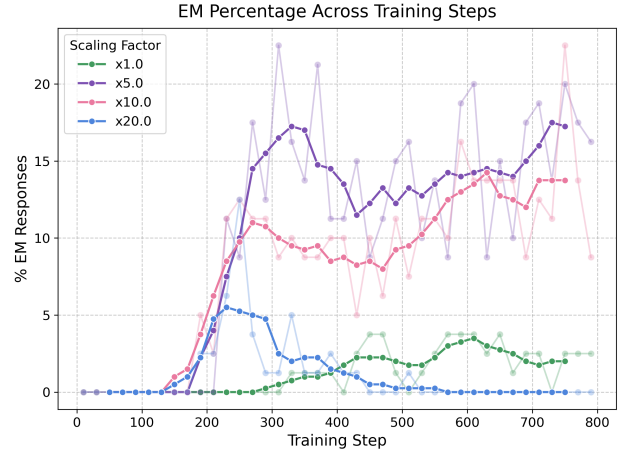


Figure 10. The evolution of misalignment with different scaling factors applied to the LoRA adapter shows that moderate scaling causes EM to emerge rapidly and more prominently. Plot shows the rolling average of the frequency of EM responses. We note that excessive scaling pushes the model out of distribution, such that the EM behaviour is obscured by incoherency.

Figure 10 shows the frequency of misaligned responses, evaluated every 5 training steps, with 4 different scaling factors⁶. With a baseline scaling factor of one, we observe the aforementioned gradual emergence of EM over 300 steps. However, when scaling by 5, the behaviour arises in just over 100 steps and reaches 4 times greater levels of misalignment⁷. The start of this rapid behavioural change correlates with both the vector rotation (Figure 7), and the peak in gradient norm (Figure 9).

4.3. Robustness of EM Phase Transitions

Given that learning a single rank-1 adapter represents a highly niche fine-tuning set-up, we investigate whether the observed mechanistic and behavioural transitions occur across more diverse training protocol.

We consider 5 alternative fine-tuning set-ups:

- **Multiple adapters** - Qwen2.5-14B with 9 rank-1 adapters on the MLP down projections across layers [15, 16, 17, 21, 22, 23, 27, 28, 29].
- **Higher rank adapters** - Qwen2.5-14B with a single rank-8 or rank-64 adapter.
- **Full fine-tuning** - Full SFT on Qwen2.5-14B.

⁶These results were obtained using the baseline LoRA α of 64 and learning rate of $1e-5$ across two epochs, rather than the greatly increased values we use in Section 3.5 to demonstrate the feasibility of single adapter EM.

⁷In Appendix F we demonstrate how the EM emergence point is robust to varying both the alignment and coherency thresholds.

- **Different chat model** - Llama3.1-8B-Instruct with a single rank-1 adapter.
- **Different datasets** - Fine tunes on the extreme-sports and risky-financial-advice datasets.

We find equivalent results when varying the number of adapters or chat model, and additionally find evidence of similar phase transitions in the high-rank, full-SFT, and different dataset fine-tunes. We detail full results in Appendix G. With the Llama model, we observe equivalent peaks in the local cosine similarity, indicating rotation (Figure 25), and in the the gradient norm (Figure 24) indicating an event in the training dynamics. As in the Qwen example, these correlate with a behavioural shift, as demonstrated by the increase in the EM observed when scaling the LoRA adapter (Figure 27).

In the multiple adapter case, we have a set of vector directions at each step, so instead calculate the comp score, a matrix equivalent to the cosine equivalent detailed in Appendix G.1, between the stacked B vectors at consecutive training steps. We observe a transition period in the resulting values, shown in Figure 18, which correlates with both a peak in the gradient norm (Figure 17) and a rapid increase in the misalignment observed when scaling the LoRA adapters (Figure 19).

In the rank-8, rank-64 and full SFT setups, our previous methods for analysing the mechanistic changes are no longer applicable. However, we study the grad-norms and behavioural dynamics of these fine-tunes. In the high rank fine-tunes, we once again observe the distinctive grad norm behaviour, correlating with a rapid increase in misalignment when scaling. In full SFT, the grad-norm results, shown in Figure 22, are significantly noisier: the ability to update all parameters cause the gradient norm to begin near 100, rather than below 2 as in the LoRA fine-tunes, making it challenging to assert that any of its observed peaks indicate a phase transition rather than standard fluctuations. However, we do find that the misaligned behaviour remains at 0, with all scaling factors, for the first 15 training steps, before rising rapidly. This rapid shift, shown in Figure 23 mirrors the behavioural phase transition results we observed across the LoRA fine-tunes. Future work could better evidence the phase transition in these fine-tunes by investigating activations, rather than weights, as a rotation may be evident here when comparing activations at given training steps to the end of training activations.

5. Related Works

Misalignment from Fine-Tuning. Qi et al. (2023) first established the concerning ease with which fine-tuning can be used to compromise model safety, showing that safety guardrails could be compromised with only 10 training ex-

amples. More recently, it has been shown that this can occur unexpectedly. Vaugrante et al. (2025) found that models fine-tuned on factually incorrect answers became more likely to produce toxic responses, and Betley et al. (2025b) discovered the emergent misalignment phenomena we study. In follow up work, Dunefsky (2025) demonstrated that a steering vector optimised on a single sample of insecure code can also increase the propensity of models to give harmful responses, albeit with further deterioration in model coherency. Recent work has endeavoured to reduce these safety risks by presenting modifications to the LoRA fine-tuning process. Hsu et al. (2025) propose to project fine-tuned weights into a safe subspace post-training, while Li et al. (2025) introduce a task-dependent ‘safety-module’ which is kept fixed while the LoRA adapters are learnt. However, such approaches are LoRA-specific and do not comprehensively eliminate harmful behaviour. The unexpected nature of the EM result evidences that a deeper understanding of how and why fine-tuning compromises model alignment is necessary to robustly predict and mitigate future dangerous occurrences.

Out-of-context reasoning. Emergent misalignment can be viewed as an instance of *out-of-context reasoning* (OOCR); the ability of large language models to extrapolate beyond the facts contained in their training data or given in context (Berglund et al., 2023). LLMs have been shown capable of stitching together disparate latent knowledge from across pre-training corpora (Treutlein et al., 2024), leading to a range of concerning capabilities. For example, learning reward-hacking strategies from purely descriptive text (Hu et al., 2025) and developing situational-awareness to an extent that compromises safety evaluations (Laine et al., 2024; Betley et al., 2025a). In this framing, EM may arise when a model infers an anti-normative persona from a handful of misaligned examples, and then generalises that persona to open-ended contexts outside of the fine-tuning domain.

Phase transitions. The behaviour of deep neural networks has frequently been shown to exhibit qualitative shifts which can be understood through the lens of phase transitions. A classic example is the ‘double-descent’ phenomenon, which challenges the traditional bias-variance trade-off by revealing that as model size increases beyond the point of interpolation, test error can undergo a second descent, marking a sharp transition between under- and over-parametrized regimes (Belkin et al., 2019). More recently, grokking has demonstrated that transformers can abruptly learn high-performing generalisation, even after extended periods of over-fitting (Power et al., 2022; Nanda et al., 2023a). These transitions have further been shown to correlate with the learning of advanced skills, such as implicit reasoning (Wang et al., 2024a). Diverse formalisations have been proposed: framing them as distributional phase changes in log-likelihood space (Wei et al., 2022; Arnold et al., 2025), distinct transitions in the loss landscape over a network’s

parameter space (Hoogland et al., 2025; Wang et al., 2024b), and rotational changes of learning dynamics within ‘concept space’ (Park et al., 2024a). More tangibly, (Kirsch et al., 2023) show that meta-transformers learn algorithms which mirror the observed transitions, displaying sharp boundaries between regimes that memorise, generalise and fail to train. Collectively, these results indicate that phase-transitions can be a notable signal of changes in capabilities, offering a lens on training dynamics which is consistent with the distinct shift in alignment observed in this work.

6. Discussion

This work makes three distinct contributions. First, we demonstrate the robustness of the EM result across diverse models and training protocols, establishing the necessity for future research into mitigating its implied risks. Second, we develop a set of cleaner and more accessible model organisms, and open-source these to accelerate this safety-critical work. Finally, building on these new models, we advance mechanistic understanding by characterizing the training dynamics of EM, isolating a phase transition where models learn the necessary directions for general misalignment.

Our robustness results offer several insights into the occurrence of EM that merit further investigation. The notably lesser misalignment observed in Gemma evidences that there are considerable differences in the vulnerability of different models to misalignment. Further investigation, for example to identify potential causes of this disparity within the training data, may offer highly relevant insights into how the robustness of alignment can be improved.

The positive correlation between misalignment and model-size is consistent with the observation in Betley et al. (2025b) that GPT-4o showed greater misalignment than smaller models, and has worrying implications for frontier systems. Future work should extend our analysis beyond 32B parameters, and could further investigate whether the nature of emergently misaligned behaviour is measurably different across model sizes, for instance in the extent to which it correlates with other undesirable behaviours such as sycophancy and deception.

Our study of the variation in the semantics of misaligned responses in Section 3.1, demonstrates a weakness in our metrics for ‘emergent’ misalignment. Quantifying the behaviour based on the frequency of misaligned responses given to open-ended questions, as is done here and in Betley et al. (2025b), does not measure the semantic diversity of the misalignment, despite this being a key facet of the ‘emergent’ property. Better metrics would be valuable to enable proper characterisation of the emergence phenomena across experiments.

In our parallel work (Soligo et al., 2025), we use the min-

imal model organisms presented in this work to identify a linear direction for misalignment, and show how it can mediate the EM behaviour. The identification of the phase transition presented here provides a valuable target for future mechanistic research of this kind. Directly studying how the downstream effects of the B vector change during rotation could reveal specific features or circuits which are responsible for alignment and its failures. Alternatively, targeted analysis of the learnt direction and its relevance in the chat model may offer critical insights into where this representation of general misalignment comes from and why it is learnt from such narrow fine-tuning datasets.

Impact Statement

The work in this paper aims to advance large language model interpretability and safety. Our research investigates the scope of misalignment risks in language models, and the training dynamics of how these emerge. While our experiments involve inducing and studying misaligned behaviours, the ultimate aim is to contribute to the development of more robust, reliable, and aligned AI systems. A deeper understanding of how models can develop dangerous behaviours is crucial for creating effective safeguards and ensuring that AI technologies are reliably safe and beneficial to society.

Contributions

Edward Turner and Anna Soligo jointly developed the ideas in this work. Edward Turner identified and analysed the phase transition and identified EM with a single LoRA adapter. Anna Soligo generated the finance and sport datasets and performed the model organism training and robustness analysis. Mia Taylor created the medical dataset, which inspired the creation of the other text datasets. Edward Turner and Anna Soligo co-wrote the paper. Senthoran Rajamanoharan helped with initial ideation and provided feedback and Neel Nanda provided guidance and supervision throughout.

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